

Color, Arousal, and Performance— A Comparison of Three Experiments

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Abstract: Three studies of the psychological and physiological effects on people of colored room interiors are described. Experiment 1 compared a colorful and a gray room, whereas in experiments 2 and 3 red and blue rooms were compared. The results indicate that the color of an interior space will have effects on many different levels. The perception of the room itself was affected, and the colors also had an impact on the emotions and physiology of those who stayed in the rooms. Strong, especially red, colors and patterns put the brain into a more excited state, sometimes to such an extent as to cause a paradoxical slowing of the heart rate. Introvert persons, as well as those already in a negative mood, became more affected than others, which caused severe changes in their performance. The series of experiments described here were among the first to be carried out in full-scale rooms painted or otherwise decorated in various colors. One practical implication is that a moderate use of good color design will serve to improve the overall mood and well-being of people. In future research more emphasis should be placed on color, as just one component in the highly complex real-life situations. © 2009 Wiley Periodicals, Inc. *Col Res Appl*, 34, 141–152, 2009; Published online in Wiley InterScience (www.interscience.wiley.com). DOI 10.1002/col.20476

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INTRODUCTION

Over the last 40 years, two groups of researchers, one in Lund and the other in Oxford, have studied the impact of colored indoor spaces on people. Three of these studies have been highly similar in terms of approach and methodology, which makes it possible to compare the outcome in some detail. The three studies specifically examine, if and how different colors will influence brain activity, mood, and performance.

Previous Research

Colors such as red, orange, and yellow are often referred to as exciting, whereas blue, blue-green, and green colors are considered to be relaxing. In the first laboratory study on the subject, in 1921, Pressey found no differences between warm and cool colors in measures of autonomic nervous response, such as pulse rate and respiration, nor in various performance tasks such as the rate of finger tapping and multiplying.¹ However, Pressey did find that brightness increased the rate of performance. Goldstein, 1942, claimed that some conditions in patients suffering from Parkinson's disease "can at times be diminished in severity if the individuals are protected against red or yellow colors, if they wear, for instance, spectacles with green lenses" (p 151).² However, others failed to replicate these findings.³ On the other hand, Schauss (1979) claimed that pink colors acted as a natural tranquilizer and had been successfully employed to reduce aggressive behavior amongst prison inmates.⁴ But once again, these results were not substantiated in later studies.⁵

The first study on the effects of color on the central and autonomic nervous systems was carried out by Gerard in 1958.⁶ In this experiment red, blue, and white lights of equal luminance were projected for 10 min on a circular translucent screen. The main finding was that the preva-

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lence of alpha waves in the electroencephalogram (EEG) was lower under red than under blue illumination (low alpha is an indicator of cortical arousal). There were also differences in blood pressure, galvanic skin response, respiration rate, and frequency of eye blinks, which were interpreted by Gerard in terms of autonomic arousal. Wilson (1966) presented his subjects with red and green slides on a translucent screen for a total of 10 min and found that the galvanic skin response was higher in the red than in the blue condition.⁷ However, the brightness of the colors had not been equated (cf. Pressey, above). In a study by Ali⁸ red or blue light equated for brightness was thrown directly into the eyes of the subject through a projector for 10 min. The results showed a greater recovery of the alpha waves of the EEG under blue than under red light, which supports Gerard's findings at the cortical level.

In a study by Jacobs and Hustmyer⁹ red, yellow, green, and blue colors of the same luminance were projected on a screen for 1 min each. There was a color effect on the galvanic skin response, but no differences in heart rate and respiration. Red was the most arousing, followed by green, yellow, and blue. Caldwell and Jones¹⁰ projected red, white, and blue light, equated for brightness, on a wall covered with white paper, and measures were taken of eye blinks, skin conductance, finger pulse volume, heart rate, and EEG activity. No color differences were found for any of the cortical and autonomic measures. The main problem with this experiment is that each presentation lasted less than 1 min.

There is some evidence that the colors of an interior space might influence the psychological mood of people in that space. Lightness, hue, and saturation might be of importance, as might the combination and distribution of the colors in the interior space. A number of studies have compared the impact of "warm" and "cool" colors (e.g., reds and blues).^{11,12} Comparing nine monochromatic office colors, Kvallek *et al.*¹³ found that saturation was a salient predictor of differences in mood between males and females. In a study by Küller *et al.*,¹⁴ of almost 1000 workspaces in four countries, interior color stood out as rather important. The mood of the workers was better throughout the year for those, who had the most colorful work environment. However, it should be understood that the majority of the environments were subdued or neutral.

Among the alleged qualities of colors is their influence on performance. Some studies seem to indicate that physical strength might increase in a red as compared to a blue environment.^{15,16} Unfortunately, room colors were poorly specified, sometimes even provided by colored light. Kvallek *et al.*^{17,18} found that the individual's screening capacity for different colors was important for their office work productivity. High screeners performed better in a red office and poorer in a blue-green office than low screeners. On the other hand, Ainsworth *et al.*¹⁹ found no differences in performance between a red and a blue-green office. It should however be noted that, whereas,

Kvallek's participants spent 4 days in the office, Ainsworth's experiment lasted for only 1 h.

Aim of the Studies

Far-reaching implications regarding the effects of colors have been derived from these studies, despite the fact that the results are hardly conclusive and that, in many instances, the color stimulation was either colored light or color patches, not real colored environments. The present article concentrates on the impact of color in indoor spaces on the general physiological state of humans, usually referred to in terms of arousal or stress reactions. Do strong colors generally activate people, or is such activation restricted to certain warm colors, like red? Do strong colors influence mood and performance in a systematic way? If so, what are the design implications of such effects? (All three studies included additional measurements, the results of which will be reported elsewhere.)

EXPERIMENT 1. AROUSAL IN MULTICOLORED AND GRAY SPACES

Problem

In the first experiment an attempt was made to bridge the gap between the divergent findings reviewed above. Why use colored light, or color patches, when instead colors could be applied in real spaces, that is, the kind of spaces we actually experience? Why test only exposures of brief duration when subjects could be exposed for much longer periods, as they normally are in everyday situations? To maximize the chances of getting results, it was decided to compare one room with an abundance of colors with another room that was completely gray. To study the possible effects we wanted to include some of the most advanced methods available at the time.

The main hypothesis was that the colorful room would be more arousing than the gray room, and that this would manifest itself both physiologically and psychologically. It was predicted that the arousal effect on the central nervous system would cause attenuation of the alpha rhythm of the EEG, which in previous research had been successfully used as an indicator of arousal. It was further predicted that there would be a corresponding effect on the autonomic nervous system revealed as an increased heart rate of the EKG. It was also assumed that there would be corresponding effects on the emotional state, or mood, of the persons staying in the colored room. Finally, it was predicted that the colorful room would be perceived to be more complex but less coherent than the gray room.

Method

The participants in the first study spent almost half a workday in one colorful and one gray room on two different occasions. (There was also one room with yellow

walls used only for introducing the participants to the procedure.) They were required to perform various tasks for a total of 3 h in each room. During this time a number of measures were taken, amongst others, EEG and EKG, and assessment of subjective mood. They were also asked to describe how they perceived the rooms by means of semantic scales.

Semantic Scales. Assessments of the perceived qualities of the rooms were made using the standardized version of the SMB form (Küller semantic environment description).^{20,21} The form was originally developed by means of factor analysis to assess the visual qualities of the built environment. It consists of 36 seven-grade rating scales giving estimates of the following eight qualities: pleasantness, complexity, unity (coherence), enclosedness, potency, social status, affection, and originality. The reported test-retest reliabilities are between $r = 0.66$ and $r = 0.86$.

Self-Reported Mood. To assess the emotional state imposed by the colors of the room a test was constructed consisting of 16 seven-grade scales representing a wide range of moods. The scales were to be answered in terms of "Right now I feel..." A factor analysis of these scales revealed four factors, which could be related to the 'basic emotional process' described by Küller in the human-environment interaction model (HEI).²¹ These emotional factors were: activation (rested/tired, awake/sleepy), orientation (interested/bored), evaluation (secure/anxious, friendly/angry), and control (talkative/quiet, strong/weak). The factor scores were then used as indices of the different moods. Later research has indicated that the coefficients α for such indices are generally satisfactory for experimental designs with repeated-measures (i.e., when participants are employed as their own controls).

Physiological Recordings. The equipment included an Elema Mingograph 800 with paper printout and a tape recorder for storage of the data. Bipolar recordings were obtained by means of electrodes (9.5 mm) placed centrally and on the parietal lobes of the left and right hemisphere (corresponding to C3-P3 and C4-P4 of the international 10-20 system), and with a baseline electrode (12 mm) on the subject's forehead. The two EEG signals were amplified by an Elema Mingograph 800, and stored on a frequency-modulated instrumentation recorder. Later, artifact-free portions of the recordings were selected and analyzed by running the tape with increased speed through an acoustic band pass filter and level recorder (Brüel and Kjaer) corresponding to the frequency range for EEG alpha (9-11 Hz).

To record EKG, an additional pair of silver electrodes (12 mm) was placed, one over the lower part of the sternum, and the other on the lateral part of the right clavicle. After amplification, the EKG was printed on paper, which was later used for estimating the pulse rate. To minimize artifacts, all cables were shielded, and the mains connections for the recording equipment were separate from the mains connections of the light sources, etc.



FIG. 1. The colorful room from Experiment 1.

Screening Interview, Measurements of Personality, Visual Acuity and Color Vision. The interview contained questions about visual acuity and color vision, incidences of unconsciousness, neurological dysfunctions, and medication. Personality was assessed with Eysenck personality inventory (EPI). Visual acuity was tested with Monoyer's chart and color vision with Tabulae Pseudo-Isochromaticae (KIFA).

Laboratory Setup. The laboratory consisted of one experimental space ($d = 430$ cm, $w = 350$ cm, $h = 247$ cm) and an adjoining control room. By changing the interior walls the former space could be transformed into three similar office rooms, one for practice and the other two for the actual study. These rooms were identically furnished with an office desk and a black chair with a cushioned seat. At the rear end of the desk there was a small gray box with an opaque window that could be illuminated by a lamp inside the box. The desk also contained some drawers for test material. At the right side of the subject there was a gray mailbox, where completed test forms could be deposited. There was a fixation point at the center of the opposing wall placed 120 cm above the floor.

In one of the experimental rooms the walls were covered with four different kinds of wallpapers with strong colors and patterns, mostly in red, yellow, and white. The floor in this room was covered with black and white tiles in a check-board pattern, and in the ceiling there were four larger, diagonally placed, boards, two white and two black. The surface of the desk was covered with multicolored stripes (Fig. 1). The other experimental room was painted in neutral gray natural color system (NCS 3000) to reduce the visual complexity. This included not only

the walls, but also the ceiling and the floor, and the top of the desk.

The rooms lacked windows and were illuminated solely with incandescent bulbs. In the colorful room the bulbs were visible through a transparent circular plastic sheet, whereas in the gray room, the bulbs were covered with a nontransparent gray board. The rooms were equated for lightness by means of a lux-meter, and the horizontal illuminance at the desktop was 40–50 lx. The practice room had the same size and was furnished and equipped in the same way as above. However, the walls were pale yellow, and the desk was in natural wood. The ambient room temperature was maintained between 22 and 24°C by means of a thermostat, and noise levels were generally very low. The participants wore clothes in neutral colors.

Procedure of Experiment 1. Each participant spent about 3.5 h in each room either in the morning, afternoon, or evening on three different occasions with 1 week in between. To get used to the procedure all participants were first exposed to a practice condition in the yellow room, and were then exposed to the colorful and gray rooms in a balanced order of presentation. The actual study lasted 3 h, when the participant was left alone in the room. An additional half hour was required for instructions and the attachment and removal of electrodes. During the actual study the research assistant was supervising the recordings in an adjoining control room, where most of the recording equipment was situated. All instructions were given in closed envelopes, which were to be opened when the light-box on the desk was momentarily turned on. The completed test forms were to be put into the mailbox. In case of emergency the participant could call the attention of the assistant by means of a push button.

During each session the participant was required to perform various tasks, either two or three times. The semantic evaluations of the room were carried out immediately before and after the actual experiment. The participants rated their emotional state after 5, 65, and 125 min. The recordings of EEG and EKG were done after 45, 105, and 165 min. During the 5-min recordings the participant was instructed to look at the fixation point at the opposite wall. Measurements of visual acuity and color vision were obtained on a separate occasion. The participants had obtained information beforehand about the general aim and procedure of the study, and participation was by informed consent.

Participants

Twelve high school and university students took part in the study, 6 males and 6 females aged 17–28 years ($m = 22.1$, $SD = 3.09$). No major medical or visual disabilities were detected.

RESULTS

The semantic ratings were carried out immediately before and after each session. Both rooms were experienced as

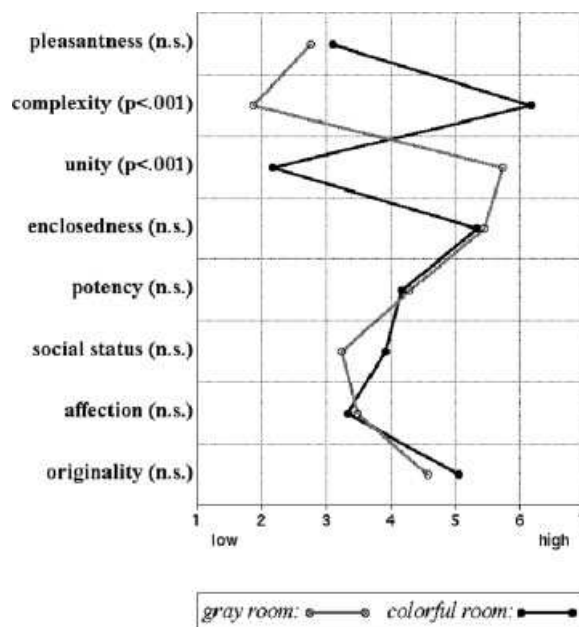


FIG. 2. Semantic profiles for the colorful and the gray room ($N = 12$).

slightly unpleasant, somewhat closed and original in character (Fig. 2). The differences between the two rooms were submitted to analysis of variance with repeated measures. The colorful room was rated to have a very high complexity, whereas the gray room obtained a very low score in this factor ($P < 0.001$). The ratings in unity (coherence) were just the opposite with low values for the colorful room and high values for the gray room ($P < 0.001$, all $N = 12$).

The participants had assessed their emotional state at three times with 1 h in between during each session. During this time activation went down considerably, especially during the first hour, but there was no difference between the colorful and the gray room. At the same time the participants felt increasingly extraverted, that is higher emotional control. Throughout, this feeling was much stronger in the gray room than in the colorful one ($P < 0.005$, $N = 12$). There were no differences between the two rooms in emotions related to orientation (attention) and evaluation.

EEG and EKG were recorded on three separate occasions, and the differences between the rooms were submitted to analysis of variance with repeated measures. The alpha rhythm of the EEG, as measured by analog power analysis, was considerably lower in the colorful room than in the gray one ($P < 0.025$, $N = 12$; Fig. 9). Furthermore, heart rate was lower in the colorful room ($P < 0.05$, $N = 12$; Fig. 10). These differences were consistent for the whole period of 3 h. Scrutiny of the data indicates that the heart rate was slowed by as much as 10% for the most introvert participants. (According to theory of personality introvert individuals are characterized by turning inwards, away from their physical and social environment.)

EXPERIMENT 2. AROUSAL IN RED AND BLUE SPACES

Problem

In the first experiment, we had found that strong colors and patterns of a room had a very different impact on people in that room compared to when the room was completely gray. This impact encompassed both psychological and physiological effects, and indicated that strong colors could cause increased arousal of the central nervous system and a paradoxical inhibition of the autonomic nervous system. Considering the results of previous studies the next step would be to compare two rooms with different hues, namely one with predominantly red and the other with predominantly blue colors. The reason for selecting red and blue instead of red and green is that the former can be equaled in terms of lightness, which, in turn, depends on the spectral absorption spectra of the cones of the human retina.

It was hypothesized that the red room would cause higher arousal in the central nervous system than the blue room as measured by EEG. Specifically, it was predicted that this would show as a blocking of the alpha waves and possibly also the delta waves. It was further hypothesized that the red room would cause an inhibition of the autonomic nervous system in terms of a slowing of the pulse rate as measured by EKG. Finally it was assumed that there would be differences in the perception of the two rooms, and that those who stayed there might be influenced as to their emotional state and performance. Possibly they would work faster in the red room compared with the blue room.

Methods

The participants in this study spent 2½ h in one red and one blue room at two different occasions. During these occasions recordings of brain and heart activity were taken, and the subjects completed various forms including assessment of their mood and a performance task, and an environmental description of the two rooms by means of semantic scales.

Semantic Scales. Assessments of the perceived qualities of the rooms were made using the standardized version of the SMB form (cf. Experiment 1).

Self-Reported Mood. The emotional state was assessed by 36 rating scales including those, which had been used in the first study. A factor analysis revealed the same four basic factors as previously obtained, namely activation, orientation, evaluation, and control (cf. Experiment 1). The factor scores were employed as indices of mood shift in the two rooms.

Measure of Performance. Performance was studied by asking the participants to write a story about what the room made them think about. They were prompted to formulate what the room reminded them of and what feelings they got. The stories were scored for number of words.

Physiological Recordings. Bipolar recordings were obtained by means of electrodes (9.5 mm) placed cen-

trally and over the parietal lobes of the left and right hemisphere (corresponding to C3–P3 and C4–P4 of the international 10–20 system), and with a baseline electrode (12 mm) on the subject's forehead. The two EEG signals were amplified by an Elema Mingograph 800, and analyzed on-line by means of analogue band pass filters in four ranges, delta (2.25–3 Hz), theta (4.5–6 Hz), alpha (9–12 Hz), and beta (18–24 Hz). To estimate mean power the filtered signals were integrated over 20-s intervals and stored on punch tape. Before inclusion in the statistical analysis the EEG-signals were inspected in order to avoid sections with eye movements and other types of artifacts.

To record EKG, an additional pair of silver electrodes (12 mm) was placed, one over the lower part of the sternum, and the other on the lateral part of the right clavicle. After amplification, the pulse rate was estimated for the same 20-s intervals. To minimize artifacts, all cables were shielded, and the mains connections for the recording equipment were separate from the mains connections of the light sources etc.

Screening Interview, and Measurements of Visual Acuity and Color Vision. The methods were similar to the ones used in Experiment 1. Visual acuity was tested with Monoyer's chart and color vision with the Ishihara plates for color deficiency (Kanehara, Tokyo).

Laboratory Setup. The experiment took place in two office mockups ($d = 293$ cm, $w = 285$ cm, $h = 300$ cm) identical except for the colors. Inside each room there was a desk (100×145 cm) with a tablecloth and a gray writing pad, and a black chair with a cushioned seat. The door was covered with a curtain. At the right side of the subject there was a small carpet and a gray mailbox, where completed forms could be deposited. There was a small print with a landscape scene on the right wall, a poster on the left wall, and an abstract oil painting on the opposing wall. A small stand with a glass decanter was located close to the right of that wall, and to the left side there was a gray loudspeaker. There was a fixation point at the center of this wall placed 120 cm above the floor.

In one test room, the walls were painted in a fairly strong red hue (NCS 1958 Y90R), and in the other room they were painted in a fairly strong blue hue (NCS 2158 B09G). The colors were equated for lightness and saturation both subjectively and by means of luminous reflectance values. The pureness of the hues was determined by means of spectral reflectance values. The components of the interior decoration, including the tablecloth, carpet, curtain and pictures, were in matching colors, whereas the ceiling was left white (NCS 0502 Y) and the floor was grayish (NCS 5000). The participants wore clothes in neutral colors.

In each room, the window ($w = 150$ cm, $h = 170$ cm) was covered with a white curtain allowing a certain amount of daylight. In addition, artificial lighting was provided by eight luminaries with square louvers, each with one fluorescent tube [Duro Test True-Lite (Fairfield, NJ) 40 W, 5500 K, CRI = 91]. Measures of room illuminance were obtained immediately before and after each session.

There were no overall differences in lighting levels between the red and the blue room. Because of the location of the rooms, illuminances were higher in the morning than in the afternoon, but this was balanced out in the experimental design. The ambient room temperature was maintained around 22°C by means of a thermostat, and noise levels were generally very low.

Procedure of Experiment 2. Each participant spent 2½ h in each room either in the morning or afternoon at two different occasions within a 2-week period. The order of presentation of the red and blue room was carefully balanced. The actual study lasted 2 h, when the participant was left alone in the room. An additional half an hour was required for instructions and the attachment and removal of electrodes. During the actual study the research assistant was supervising the recordings in an adjoining control room, where most of the recording equipment was situated. Instructions came via the loudspeaker from a tape recorder, and the completed test forms were to be put into the mailbox. In case of emergency the participant could call the attention of the assistant.

Each session included a number of tasks, some of which were carried out twice. The semantic evaluations of the room were carried out after 5 min and again after 80 min. The participants rated their emotional state after 30 and 100 min. The recordings of EEG and EKG were made after 40 min and after 110 min. During the 5-min recordings the participant was asked to look at the fixation point at the opposite wall. The performance task commenced after 60 min and lasted for 15 min. Measurements of visual acuity and color vision were obtained on a separate occasion. The participants had obtained information beforehand about the general aim and procedure of the study. Participation was by informed consent.

Participants

Twenty-five students from the School of Architecture were asked to take part in the study (8 males and 17 females aged 20–31 years; $m = 23.7$; $SD = 3.02$). Because of major color deficiency one participant was excluded from all further data treatment. A few others had to be partially discarded due to incomplete data.

RESULTS

The semantic ratings were carried out twice during each session. The differences between the two rooms were submitted to analysis of variance with repeated measures. As shown in Fig. 3 both rooms were about average in pleasantness, complexity, unity (coherence), affection, and originality, and both were low in social status. The red room was perceived as more closed ($P = 0.02$) and to have higher affection ($P = 0.008$) than the blue one, whereas the blue room was considered to have higher potency ($P = 0.004$) and somewhat higher social status, relatively speaking ($P = 0.01$, all $N = 23$).

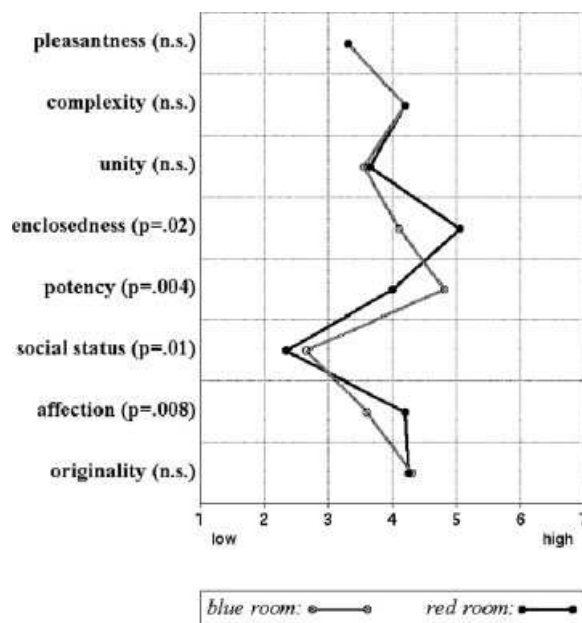


FIG. 3. Semantic profiles for the red and the blue room in experiment 2 ($N = 23$).

The participants assessed their emotional state two times during each session in terms of activation, orientation, evaluation, and control. During this time they became somewhat more awake and, at the same time, slightly bored. However, there were no over all differences between the red and the blue room in any of the four emotional dimensions.

A majority of the participants (15 as compared to 7) wrote longer stories in the red room than in the blue room. However, this difference did not reach full statistical significance ($P = 0.07$, $N = 22$).

EEG and EKG were recorded on two separate occasions, and the differences between the rooms were submitted to analysis of variance with repeated measures. Concerning EEG, two of the frequencies (δ and α) showed significant or near significant differences. Delta waves characterizing a sleepy and drowsy state were found in greater abundance in the blue condition ($P = 0.02$, $N = 20$). Alpha rhythm was also higher in the blue room, even if this was only a tendency ($P = 0.16$, $N = 20$; Fig. 9). During the second hour this difference increased, especially over the left hemisphere of the brain ($P = 0.11$, $N = 20$). Frequencies θ and β showed no differences between the two rooms. Heart rate was consistently lower in the red room than in the blue room ($P = 0.03$, $N = 18$; Fig. 10).

EXPERIMENT 3. PERFORMANCE IN RED AND BLUE SPACES

Problem

Mood and work efficiency might be related to the hedonic and arousing qualities of a workspace. Some environments can be overly stimulating, whereas others

can be boring and monotonous, and this might influence the persons working there. Such effects have been demonstrated in the social realm. For instance Zajonc found that the presence of others enhanced the productivity on simple routine tasks but that the productivity on complex tasks decreased.²² This result was corroborated in a similar study by Block and Stokes.²³ Is it possible that colors in the workspace might have similar effects?

The purpose of the third study was to examine the impact of room colors on the performance of two types of office work, namely routine clerical work on one hand and creative work on the other. It was hypothesized that proof reading would be accomplished more efficiently in a room with warm red colors. Further it was hypothesized that essay writing would be more creative in a room with cool blue colors. It was also assumed that these effects might be mediated by the individual's mood or emotional status. Finally, it was predicted that there would be differences in the perception of the two rooms similar to the ones obtained in experiment 2.

Method

Semantic Scales. Assessments of the perceived qualities of the rooms were made using the standardised version of the SMB test (cf. Experiment 1).

Self-Reported Mood. Based on the HEI model,²¹ mood was assessed on four-grade rating scales in terms of four emotional qualities: activation (rested/tired, alert/drowsy, awake/sleepy), orientation (interested/bored, efficient/inefficient, devoted/indifferent), evaluation (secure/anxious, friendly/angry, happy/sad), and control (confident/hesitating, independent/dependent, strong/weak). (This was a development of the scales employed in the first and second experiment. When the scales are combined into a total index of emotional status, coefficient $\alpha = 0.82$).¹⁴

Measures of Performance and Creativity. Routine performance was assessed by proof reading and correcting a lengthy text consisting of chapters from the official Swedish Law Book, into which spelling errors had been randomly entered, but at the general rate of one error per one hundred signs. The instruction was to work as fast as possible for 3 h. The accumulated corrections were automatically registered every tenth minute.

Creativity was assessed by letting the participants write an essay with a content, form and length of their own choice. As an inspiration, each session was introduced by showing five black-and-white pictures of figurative paintings. The instruction was to work for three full hours. When 10 min remained a notice was displayed to facilitate the completion of the composition. The accumulated production was automatically registered every tenth minute.

Screening Interview, and Measurements of Visual Acuity and Color Vision. The methods were similar to the ones used in experiment 1 and 2. In addition, visual performance was assessed by means of a reading test on the



FIG. 4. The red office from Experiment 3.

visual display unit consisting of seven texts varying in font size from 4 to 20.

Laboratory Setup. The laboratory was arranged as three similar office rooms, one for practice and the other two for the actual study. These two rooms ($d = 330$ cm, $w = 286$ cm, $h = 300$ cm) were identically furnished with an office desk and a computer, two chairs, a small sideboard, two pictures on the walls and a waste-paper basket (Fig. 4). The ceilings and the upper parts of the walls (50 cm) were left white (NCS 0502 Y) and the floor was grayish (NCS 5000). In one test room the walls were painted in a fairly strong red hue (NCS 1360 Y90R), and in the other room they were painted in a fairly strong blue hue (NCS 1355 B10G). The colors were equated for lightness and saturation both subjectively and by means of luminous reflectance values. The pureness of the hues was determined by means of spectral reflectance values. The participants wore clothes of their own choice.

In each room, the window ($w = 150$ cm, $h = 170$ cm) was completely covered by a heat reflecting blind and a curtain in a color corresponding to the color of the walls. Lighting was provided solely by artificial illumination consisting of four luminaries with square louvers, each with three fluorescent tubes (Duro Test True-Lite (Fairfield, NJ) 40 W, 5500 K, CRI = 91) powered by high-frequency ballasts. The two luminaries farthest away from the participant were screened to eliminate possible glare. The horizontal illuminance at the desktop was 500 lx. The computer hardware consisted of Macintosh LCII computers with low noise levels and 15" calibrated black-and-white screens (120 candela/m²). The practice room had nearly the same size and was furnished and equipped in the same way as above. However, the walls were white (NCS 0502 Y) and so was the curtain. The temperature in each room was measured before and after each session with a digital thermometer ($m = 22.8^\circ\text{C}$, $SD = 1.25$). Noise levels were very low throughout the sessions.

Procedure of Experiment 3. For practical reasons the semantic evaluations of the two test rooms were carried out in a pilot study by a separate group of persons. In the main study half of the participants, matched for age and

gender, were assigned to the proof reading task and the other half was assigned to the creative task. Each participant spent about 3.5 h in the laboratory office on three separate occasions, either in the morning or in the afternoon, with 1 week in between. All assignments and assessments were carried out in both the red and the blue room in a balanced order of presentation. To get used to the procedure all participants were first exposed to a practice condition in the white room. After a brief introduction the participant was left alone in the room.

Each session began by letting the participant assess her or his emotional state. Five minutes later the main task started. For half of the participants this consisted of proof reading, whereas the other half began to write an essay. This task then went on for 3 h. Finally, after another 10 min, the emotional state was assessed a second time. Apart from the screening of visual acuity and color deficiency, which took part on a separate occasion, all tests in the pilot study and the main study were presented and answered directly on the computer screen. The participants of the main study had obtained information beforehand about the general aim and procedure of the study without specifically mentioning the color effects. Participation was by informed consent, and adhered to the ethical standards set down by the Swedish Psychological Association.

Participants

Twenty university students took part in the pilot study, 8 males and 12 females aged 21–54 years ($m = 32.4$, $SD = 7.6$). Forty persons with a wide range of occupations participated in the main study, 20 males and 20 females, between the ages of 21 and 62 years ($m = 40.9$, $SD = 13.2$). Twenty-five wore eyeglasses or contact lenses. No major medical or visual disabilities were detected.

RESULTS

As aforementioned, the semantic ratings of the two office rooms were carried out in a pilot study. The differences between the two rooms were submitted to analysis of variance with repeated measures. As shown in Fig. 5 both rooms were about average in complexity, somewhat high in unity (coherence), enclosedness and originality, but somewhat low in social status and affection. In the remaining two factors there were significant differences between the two rooms. The red office was perceived as more pleasant than the blue one ($P < 0.01$), whereas the blue office was considered to be higher in potency ($P < 0.001$, all $N = 20$).

At the beginning and end of the 3-h session the participants of the main study had assessed their emotional state in terms of activation, orientation, evaluation, and control. During this time activation and orientation went down considerably, that is, they felt more tired and bored, but there was no difference between the red and blue rooms in these respects. However, the participants evaluations were somewhat more positive in the red room than in the

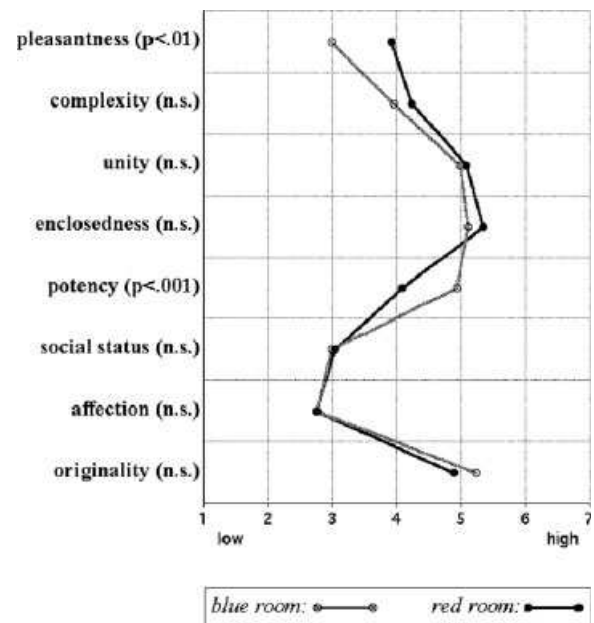


FIG. 5. Semantic profiles for the red and the blue office in experiment 3 ($N = 20$).

blue room, that is, they felt somewhat more happy in the red room throughout the whole session ($P = 0.01$, $N = 40$). There was no difference between the two rooms in emotions related to emotional control.

Measures of performance and creativity had been carried out with two subgroups of equal size. Performance was measured as the total amount of handled text (number of signs) and percentage of overlooked errors. These two measures were highly correlated ($r = 0.72$, $P < 0.001$, $N = 20$), but there was no overall difference in performance between the two office rooms. Although most participants upheld a remarkably steady pace during the 3-h work session, with an average of 481 signs per minute and 21% errors, there were very large individual differences that warranted a search for group patterns. This revealed that the younger participants handled significantly more text than the older ones ($P = 0.04$, $N = 20$).

However, the most striking difference was observed when the participants were partitioned according to their emotional state. In accordance with previous research the subscales of the emotional qualities were combined into a total index of emotional status. The participants were then divided into two groups of equal size, those with a more positive mood, and those with a more negative mood, relatively speaking. As shown in Figs. 6 and 7, there were significant differences in performance between these two groups in the red office, but not in the blue one. Participants, who were in a negative mood, checked considerably longer texts and also had a larger percentage of errors when working in the red office (Interactions: $P = 0.02$, $P = 0.06$, $N = 20$).

Creativity was assessed by counting the total length of the essay (number of signs), and by calculating the actual word length (using longer words is supposed to be more creative). Furthermore, five skilled readers compared the

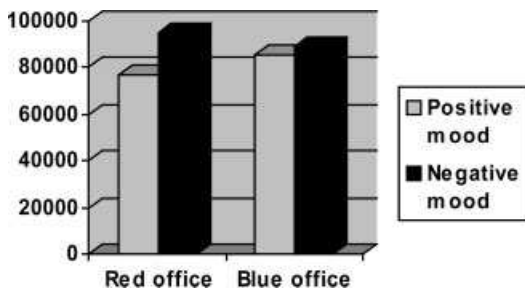


FIG. 6. Length of text (signs) corrected in the red and the blue office by two subgroups of participants, those in a high, and those in a low emotional state (Interaction: $P = 0.02$, $N = 20$).

two compositions of each participant in terms of originality, language, disposition, and entertainment value. (The concordance between the judges for the subjective measures was acceptable). However, none of the measures revealed any difference in creativity between the two offices. Once again the participants were partitioned into those with a more positive mood, and those with a more negative mood (cf. above). As shown in Fig. 8, participants, who were in a negative mood, wrote longer essays especially in the red office (Interaction: $P = 0.05$, $N = 20$).

DISCUSSION

In contrast to most previous color studies this series of recently concluded experiments took place in full scale spaces as close to real environments as was permitted under laboratory conditions. Apart from employing somewhat strong colors and patterns, care was taken to make the environments look like ordinary rooms or offices. Also in contrast to most previous studies the duration of each session lasted for several hours and covered different periods of the day. All in all a broad spectrum of participants was used including both genders, and a diversity of ages and occupations. Finally, the selection of the colors and the recordings of the participants' responses included a number of subjective, physiological, and physical methods. Undoubtedly, this has helped illuminate some of the issues of color research related to the impact of color on people. As the reader may have noticed, there is consider-

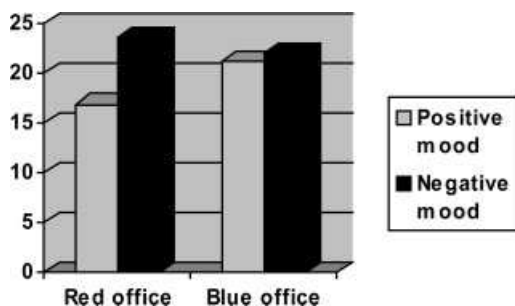


FIG. 7. Percentage of uncorrected errors in the red and the blue office by two subgroups of participants, those in a high, and those in a low emotional state (Interaction: $P = 0.06$, $N = 20$).

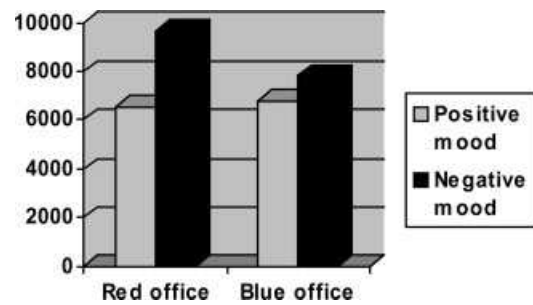


FIG. 8. Length of essays produced in the red and the blue office by two subgroups of participants, those in a high, and those in a low emotional state (Interaction: $P = 0.05$, $N = 20$).

able consistency between the different studies. However, there is also some inconsistency, and we will now try to summarize the main outcome in terms of each category of measurements.

The Perception of the Rooms

The colorful and gray rooms used in the first experiment were both perceived as somewhat unpleasant, closed, and original in character. They only differed significantly in terms of complexity and unity (coherence). In an early study by Acking and Küller (1972) semantic scales of the same type as used in the present experiments were employed to study almost 100 colored varieties of the same room, presented either as color slides of sketches or as full scale mock-ups.²⁴ Amongst the most clear-cut results of that study were that the higher the chromaticity of the interior walls, the higher became the perceived complexity, and the lower the perceived unity (coherence) of the whole room. This is fully in line with the results of experiment 1.

As for the red and blue rooms in experiments 2 and 3, the outcome is not fully consistent. In experiment 2 the pleasantness of both spaces was somewhat below average but, in experiment 3, the blue space was significantly less pleasant than the red one. In both experiments the red space was perceived as somewhat more enclosed than the blue space, however the difference became significant only in experiment 2. There were also differences in social status and affection in experiment 2 that did not reappear in experiment 3. The only result that seems completely consistent is that the blue space was experienced as higher in potency than the red one. Küller defines potency as "an expression of power in the environment" (p 132).²¹ We may conclude that, although the walls and decorations in both experiments were held in fairly strong reds or blues, this was not enough to determine how the rooms were to be perceived in every respect. It is more than likely that other design features were of considerable importance as well.

The Self-Reported Emotions

In all three experiments, the participants repeatedly assessed their mood, which makes it possible to study

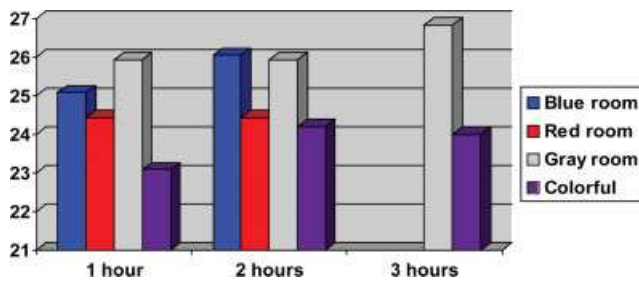


FIG. 9. The alpha rhythm (%) of the participants in experiment 1 and 2 after staying in the room for one, two, and 3 h. Low alpha rhythm indicates high arousal.

possible changes over time in the different rooms. The scales were partly identical, which enables us to compare the studies in terms of the four basic emotions described by Küller.²¹ In experiment 1 activation, and in experiment 3 activation and orientation, went down considerably, that is, the participants felt more tired, or more tired and bored, as time passed by. This is understandable, since each session lasted about 3 h. The results of experiment 2 agree with this as for orientation, but in that study activation increased somewhat. This might possibly be ascribed to the presence of daylight and the shorter duration of the study. There were no general differences in activation and orientation between the gray and the colorful room or between the red and blue rooms.

Concerning evaluation, there were no differences between the colorful and gray room. As for the two studies of red and blue rooms, there was no general difference in experiment 2, but in experiment 3 the participants felt more positive in the red room than in the blue one. This coincides with the fact that the red space was experienced as more pleasant than the blue one in experiment 3 but not in experiment 2 (cf. above).

Finally, the feeling of emotional control was much stronger in the gray room than in the colorful one throughout the whole session. This lends support to the hypothesis of experiment 1 that the colorful room would be more psychologically arousing than the gray one, which obviously resulted in withdrawal and loss of control. As far as we know, the only comparable studies on the impact of room color on mood are those by Kwallek *et al.*^{13,17} However, the experimental design of these studies prevents any direct comparison of the results.

Analysis of Brain Activity

One of the physiological methods employed in the first and second experiments was the recording and analysis of brain activity by means of electroencephalography. The EEG mainly represents the sum of the dendritic activity of thousands of nerve cells in the underlying cerebral cortex. It can be considered as a rather general measure of the brain's level of excitation.²⁵ Several principal EEG rhythms have been described in routine EEG recordings, two of which will be discussed here. Delta rhythm (<4

Hz) is characteristic of deep sleep, but may also indicate drowsiness in a person, who is awake. There is considerable evidence that alpha rhythm (8–13 Hz), especially in the posterior part of the brain, is inversely related to arousal.^{25–27}

The results of experiment 1 showed that there was considerably less alpha rhythm in the colorful room than in the gray one, thus indicating higher arousal in response to high visual complexity. This indicates that colors and patterns stimulate the brain's electrical activity. In experiment 2 the delta rhythm characteristic of a sleepy and drowsy state was found in greater abundance in the blue room, and the alpha rhythm also became higher in this room, although for alpha the difference did not reach full statistical significance. Still, taken together this provides support for the hypothesis that red interiors are more arousing than blue ones. Combining the results of these two experiments may suggest that the colorful room was more arousing than the red one, which in its turn was more arousing than the blue and the gray room (Fig. 9).

Autonomic Heart Rate Responses

The other physiological method employed in the first and second experiments was the recording of the participants' heart rate (EKG). In experiment 1 heart rate was found to be lower in the colorful room than in the gray one, which was contrary to our hypothesis. Based on the work by Lacey and Lacey, Duffy, and Libby *et al.*, an alternative explanation was formulated, namely that in environments that elicit attention, cortical arousal may be accompanied by cardiac deceleration.^{28–31} This assumption has been further substantiated in later studies.³²

Detailed inspection of the data revealed that the heart rate was slowed by as much as 10% for the most introvert participants. This is in line with Hans Eysenck's claim that extraverts have lower levels of arousal than introverts, and indicates that introverts are more susceptible to external stimulation.³³ The interpretation of heart rate inhibition as a compensatory response to visual over-stimulation was followed up in experiment 2, where it was hypothesized that the red room would cause a slowing of the heart rate. This assumption was fully supported by the results of the second experiment (Fig. 10).

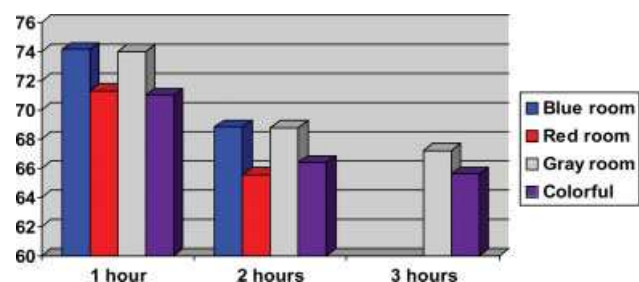


FIG. 10. Heart rate (beats/minute) of the participants in experiment 1 and 2 after staying in the room for one, two, and 3 h. Low heart rate can indicate mental attention and overload.

Performance and Creativity in Rooms with Different Colors

In experiment 2 an attempt was made to measure performance by letting the participants write stories, which were then scored for number of words. Although a majority wrote longer stories in the red room than in the blue one, statistically this difference did not reach full significance. The idea that color might have an effect on performance was further elaborated in experiment 3, where it was hypothesized that a monotonous task, like proof reading, would be more efficiently accomplished in the red office, whereas a creative task, such as writing an essay, would be favored in the blue office.

Even if no overall difference in performance between the two offices was found, there was a highly interesting interaction between room color on one hand, and the emotional state of the participants on the other. Participants, who were in a negative mood, completed more proof reading in the red office, but also missed out a larger percentage of corrections. The combination of increased speed and decreased accuracy is a well-established indicator of heightened arousal in the central nervous system.²⁶

The results by Kvallek *et al.*, support the existence of individual differences in response to colors.^{17,18} Comparing three offices with different colors, these authors found that participants, who were classified as 'high screeners', performed better in the red office and poorer in the blue office than those, who were classified as 'low screeners'. According to Mehrabian, who developed the Stimulus Screening Ability Questionnaire, 'high screeners' will score low on 'trait arousability'.³⁴ Our own results do not seem to be fully compatible with those of Kvallek *et al.*, and more research will be needed to elucidate why some persons are more sensitive to colors than others.

Creativity was assessed in several different ways, yet no effect of room color was found. However, once again those participants, who were in a negative mood, wrote longer essays, especially in the red office. It should be noted that the group, who wrote essays, was not the same as the one, who did proof reading. Naturally, the similarity of the outcomes from experiments 2 and 3 in this respect strengthens the case for an impact of color on performance.

CONCLUSIONS

This series of three experiments has clearly demonstrated that the color of an interior space will have effects on many different levels. The perception of the room itself will be affected. Significant differences were found in seven of the eight visual qualities assessed by the semantic scales. But the influence goes even deeper. The colors actually had an impact on the emotions and physiology of those who stayed in the rooms. Strong, especially red, colors and patterns put the brain into a more excited state, sometimes to such an extent as to cause a slowing of the

heart rate. Introvert persons, as well as those already in a negative mood, became more affected than others, which caused severe changes in their performance. The results warrant the conclusion that the color and complexity of the visual field has a much deeper influence upon people than has been supposed so far.

One can only speculate about the rationale behind these findings. Some of the effects must certainly be the outcome of evolutionary processes, whereas others might be cultural or a mixture of the two. The evolutionary biologist, Nicholas Humphrey, has pointed to the fact that red by far is the most common color signal in nature.³⁵ It stands out very well against a background of green foliage or blue sky. But the message it carries is rather ambiguous, because red is the color of edible and poisonous things, and also of blood. On the symbolic level red is also the color of love and anger. More than 20 years ago Peter Kaiser, in this journal, raised the question whether the physiological responses to color are cognitively mediated.³⁶ In the near future neuroscientists, employing new techniques of looking into the living brain, are likely to come up with an answer to this question.

At present, most design implications regarding colors have been derived from studies where the color stimulation consisted of colored light, color slides, or color patches. The series of experiments described here were among the first to be carried out in full-scale rooms painted or otherwise decorated in various colors. In this respect, the implications of these studies may be more directly relevant for application purposes. One practical implication would be to employ color design to accomplish minor adjustments in the general arousal level of individuals. Perhaps warm colors could be applied to add some extra arousal to an otherwise monotonous situation. However, the use of color for these purposes must never be extreme. It is not advisable, for example, to employ color to compensate for a boring work situation. Still, results from recent studies suggest that a moderate use of good color design does improve the overall mood and well-being of people.¹⁴

In future research more emphasis should be placed on color, as just one component in the highly complex real-life situations. We have reason to assume that color interacts with other characteristics of the physical environment and influences a wide range of activities, including social life. There is also some evidence of individual and cultural differences with regard to color. Undoubtedly, there are exciting studies waiting to be carried out. To ensure ecological validity, a variety of existing work environments could be photographed and made subject to perceptual studies by means of computer simulation. The next step would be to conduct long-term field studies in real environments looking at building- and work-related symptoms among the participants not only as a function of color but also of lighting, noise, heat, and other environmental factors. Such field studies could be supplemented with laboratory experiments of briefer duration, where certain responses could be further investigated by means

of psychological and physiological techniques. No doubt, such an ambitious undertaking would require the cooperation of several research groups, perhaps even from different countries.

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